

Integrated Water Vapor Field and Multiscale Variations over China from GPS Measurements

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ABSTRACT

Water vapor plays a key role in the global hydrologic cycle and in climatic change. However, the distribution and variability of water vapor in the troposphere are not understood well—in particular, in China with the complex Tibetan Plateau and the influence of the Asian and Pacific monsoons. In this paper, continuous global positioning system (GPS) observations for 2004–07 in China are used to produce precipitable water vapor (PWV) measurements; these measurements constitute the first investigation of PWV distribution and variability over China. It has been found that the stronger water vapor values are in southeastern China and the lower water vapor values are in northwestern China. These distributions are mainly affected by the latitude, topographical features, the season, and the monsoon. Water vapor variations over China are mainly dominated by seasonal variations. The strong seasonal cycles are in summer with maximum water vapor and in winter with minimum water vapor. The PWV in southeastern China has an annual amplitude of about 15 mm, much larger than in northwestern China at about 4 mm, and meanwhile the time of peak water vapor content is one month earlier than in other regions, probably because of the known rainy season (mei-yu). In addition, significant diurnal variations of water vapor are found over all GPS stations, with a mean amplitude of about 0.7 mm, and the peak value occurs around noon or midnight, depending on geographic location and topographical features. The semidiurnal cycle is weaker, with a mean amplitude of about 0.3 mm, and the first peak PWV value appears around noon.

1. Introduction

Atmospheric water vapor is a variable that interacts with the solar radiation and controls the thermodynamics and energy balance of the atmosphere. Therefore, water vapor plays a key role in the global hydrologic cycle and heat processes of the climate system. Integrated or precipitable water vapor (PWV) is an impor-

tant indicator of water vapor variability in the lower troposphere and related climate processes. It represents the water vapor storage in the column of the atmosphere in terms of water budget, as a result of the balance among precipitation, evaporation, and convergence of humidity (Fontaine et al. 2003). PWV concentrations possess a high degree of spatial and temporal variability that depends on the season, topographic features, and other local and regional climatic conditions, especially in China with the highest Tibetan Plateau and being subject to the Asian and Pacific monsoons. However, the spatial and temporal distribution features of water vapor concentrations were not understood well until radiosonde data became available (Bannon

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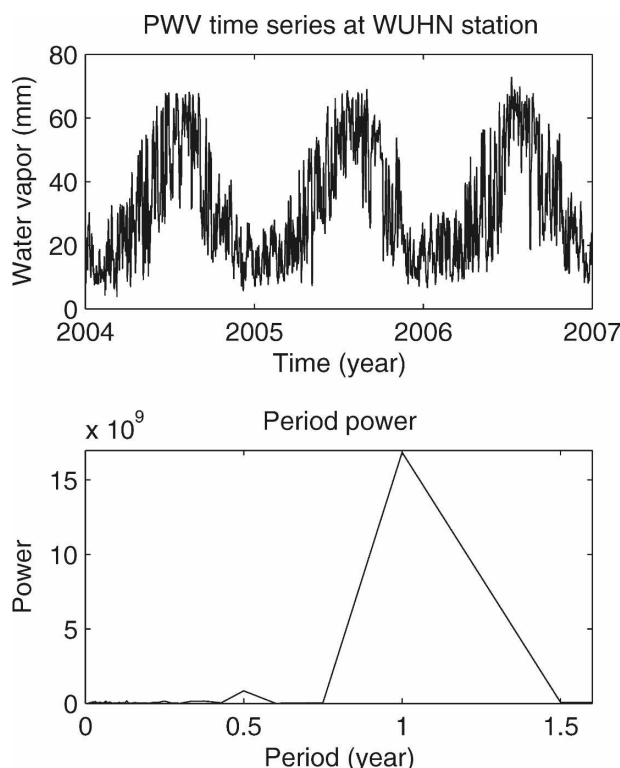


FIG. 2. (top) PWV time series at WUHN station for 2004–07 and (bottom) period power chart.

observations from the Comprehensive Ocean–Atmosphere Dataset (COADS; <http://dss.ucar.edu/datasets/ds464.0>) and the absent surface temperatures are estimated from the 6-hourly National Centers for Environmental Prediction–National Center for Atmospheric Research reanalysis products (<ftp://ftp.cdc.noaa.gov/Datasets/ncep.reanalysis>; Wang et al. 2005). Thus, 2-h PWV time series for 2004–07 over China can be obtained from GPS-derived ZTD and surface pressure and temperature data, and the uncertainty of PWV is about 0.5–1.5 mm because of errors in ZTD, surface pressure, and temperature. For example, Fig. 2 shows the water vapor time series at the Wuhan, China, (WUHN) station for 2004–07. In general, the GPS-derived PWV agrees well with that from radiosonde in Beijing and Shanghai, China, and with collocated very long baseline interferometry (VLBI) observations in Shanghai, with a mean difference of about 1.8 mm and mean root-mean-square (rms) of about 1.5 mm.

3. Results and discussion

a. Water vapor distribution

Precipitable water vapor is a key parameter of the atmosphere and varies with season, topographic features, and other climatic conditions. The 3-yr PWV

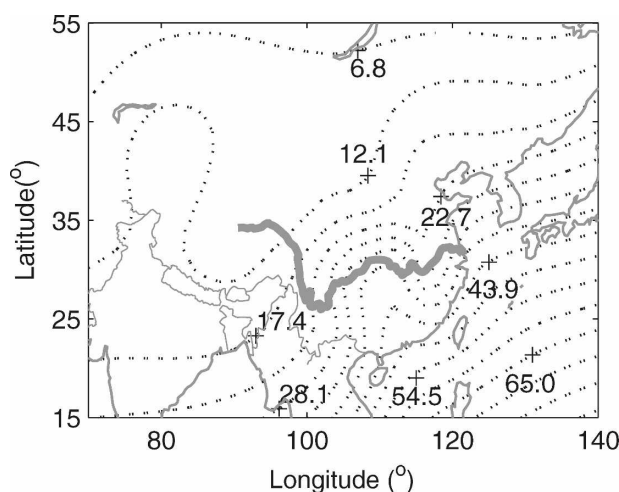


FIG. 3. Contour distribution of mean PWV (mm) over China. The mean PWV at each site was calculated from the 2-h PWV time series over 3 yr. The thick line in the middle is the Yangtze River valley.

time series with 2-h sampling at 28 GPS sites over China provide an important data source to investigate water vapor distribution features and to understand its variability and change. The mean PWV was calculated from the 2-h PWV time series over 3 yr with the sample size of about 12 000 at each station. Figure 3 shows the distribution of mean PWV as a contour with kriging interpolation (Isaaks and Srivastava 1989). Over northern and western China the water vapor content is lower, ranging from 6.8 to 20.0 mm, and over the Yangtze River valley (shown by the thick line in the middle part of Fig. 3) and southeast China it is about 40.0–65.0 mm. It can be seen that the water vapor distribution over China has a latitudinal feature, as typified in the east of

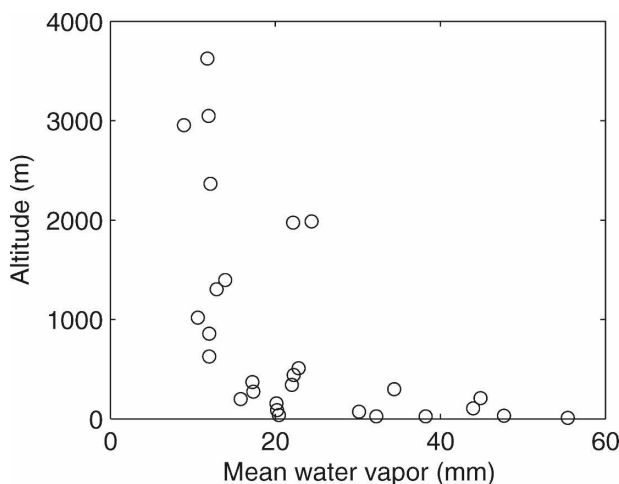


FIG. 4. Distribution of mean PWV with altitude (above global mean sea level). The circles represent the mean PWV at each site.

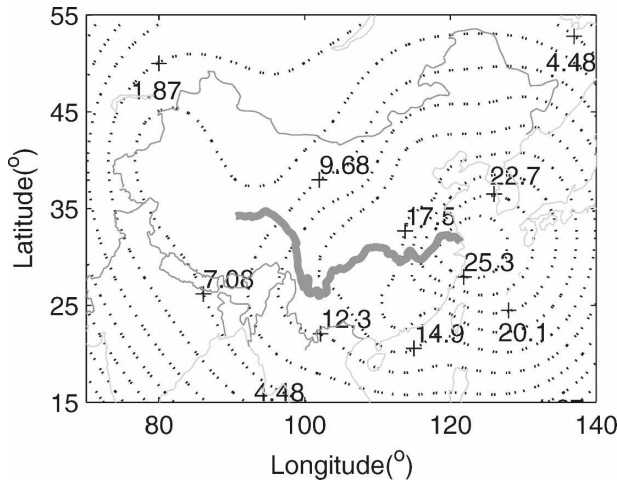


FIG. 5. Contour distributions of annual PWV amplitudes over China.

China. As the latitude increases, the water vapor amount decreases, with an obvious difference between southern China and northern China. The distribution features of PWV closely match China's geographical locations and probably reflect the influence of the Asian and Pacific monsoons (Chang 2004). Higher PWV in the southeast of Tibet is due to water vapor transportation from the Bay of Bengal and adjacent areas (Zou et al. 1990). At the same latitude, the water vapor over western China is lower than over eastern China, which may be caused by a colder atmosphere over the high, snow-covered surface over the western areas (e.g., Tibet). Figure 4 shows the distribution of mean PWV with altitude (above global mean sea level). The circle represents the mean PWV at each site, with

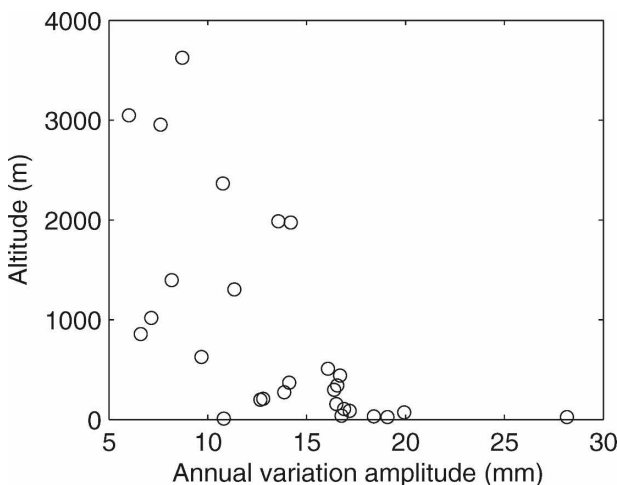


FIG. 6. Distribution of annual PWV variation amplitudes with altitude. The circles represent annual PWV variation amplitude at each site.

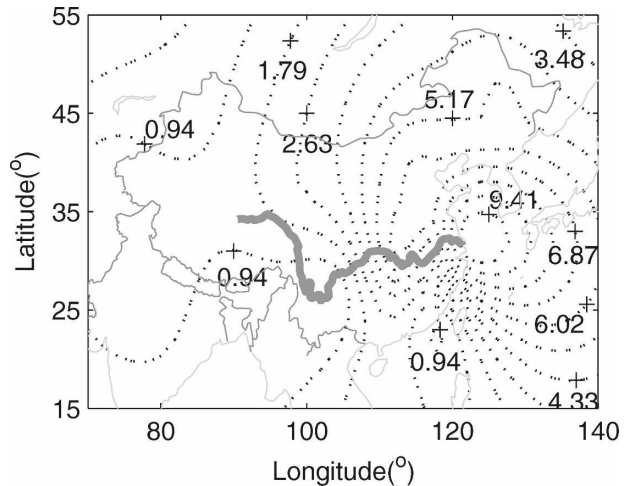


FIG. 7. Contour distributions of semiannual PWV amplitudes over China.

uncertainty of 0.5–1.5 mm. An obvious descent of the water vapor content is seen with increasing elevation. Therefore, the distribution of atmospheric water vapor over China is mainly dominated by the latitude, topographical features, and climatic conditions. In addition, because of nonuniform GPS station distribution—in particular, in western China—the interpolated water vapor distribution features need to be further investigated with future denser observations.

b. Seasonal variations of water vapor

We analyzed all of the PWV time series at each GPS station with fast Fourier transform and found that the most marked periods of PWV time series at all GPS

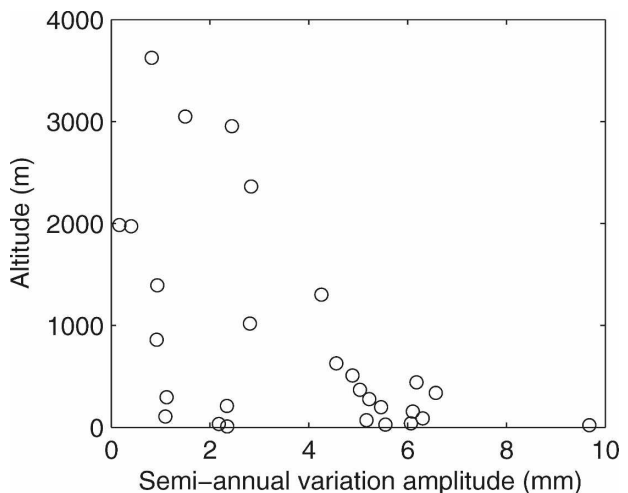


FIG. 8. Distribution of semiannual PWV variation amplitudes with altitude. The circles represent semiannual PWV variation amplitude at each site.

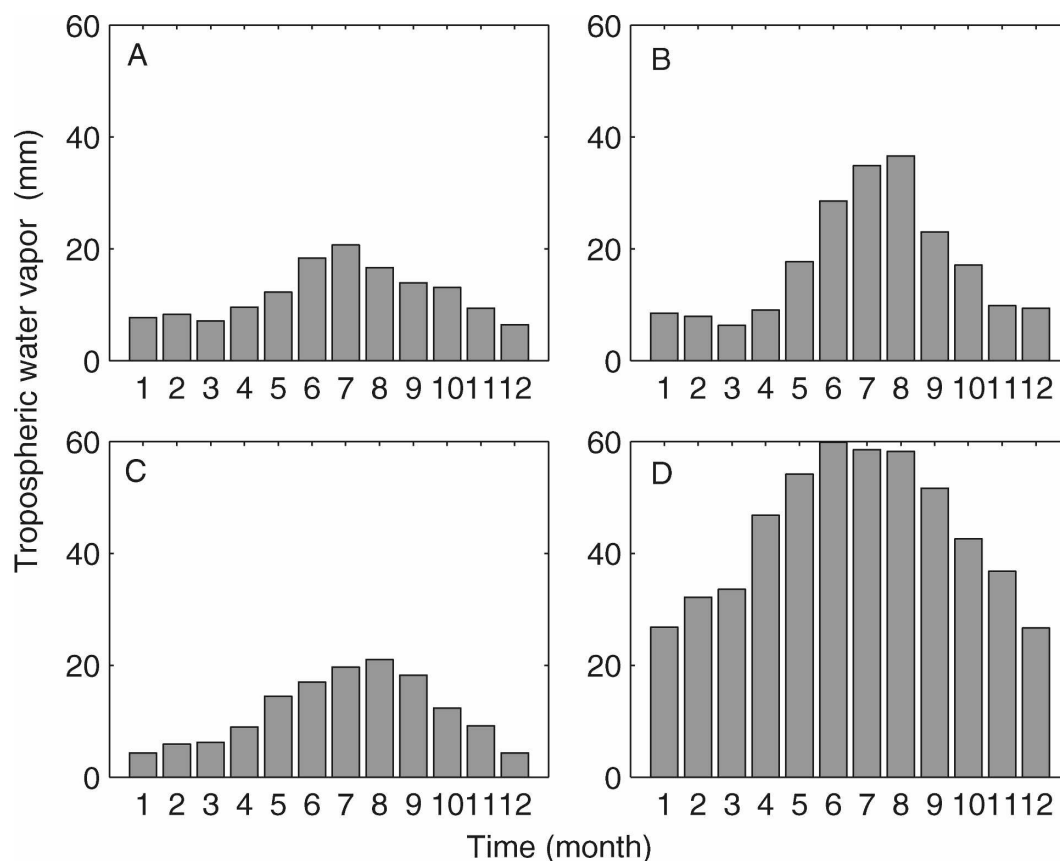


FIG. 9. Distribution of monthly PWV at the four subregions (A–D) marked in Fig. 1.

stations are about 1 and 0.5 yr. For example, Fig. 2 (bottom panel) shows the period power chart of PWV time series at the WUHN GPS site. The strong 1-yr and weak 0.5-yr periods have shown clear significant annual and weak semiannual variations, respectively. The characteristics of annual and semiannual terms at each site are exacted from 3-yr PWV time series. The uncertainty is about 0.7 and 0.4 mm for annual and semiannual amplitudes, respectively. The strong seasonal cycles are in summer with a maximum water vapor because of the influence of a moist summer monsoon and in winter with a minimum water vapor because of cold temperatures and dry weather. Figure 5 shows the amplitude of annual water vapor variations over China. The higher amplitudes are found in mid-eastern China with about 25.3 mm, and the lower amplitudes are focused on western China. The annual-variation amplitudes are closely related to the topographic elevation (see Fig. 6). The semiannual-variation amplitudes are weak, with approximately several millimeters, much lower than annual variations—in particular, in western China with insignificant semiannual variations (Fig. 7) that are also related to the topographic elevation (see Fig. 8).

Based on the topographic features, China is divided into four subregions in Fig. 1 marked as A (northwestern China), B (northeastern China), C (southwestern China), and D (southeastern China). The mean monthly medians of each subregion are obtained from 3-yr PWV time series and are shown in Fig. 9. It can be

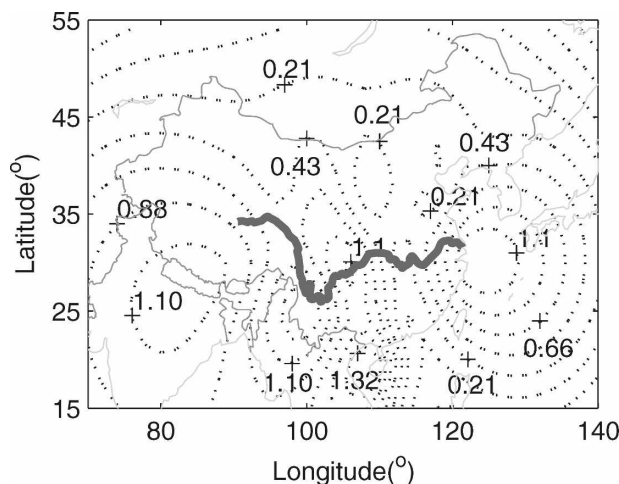


FIG. 10. Contour distributions of diurnal PWV amplitudes over China.

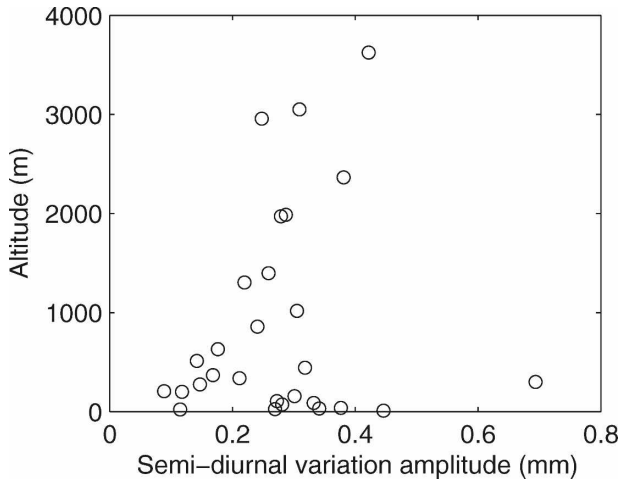


FIG. 11. Distribution of semidiurnal PWV variation amplitudes with altitude. The circles represent semidiurnal PWV variation amplitude at each site.

seen that the higher monthly water vapors are in eastern China (B and D) while western China (A and C) has lower monthly water vapors, but the entire water vapor in each region has a significant seasonal variation. The peak PWV content appears in July–August,

and the minimum PWV value occurs in January in southeastern China. However, the peak PWV values in the western China regions (A and C) are much lower than the minimum value in southeastern China (D). The peak time also has an anomaly in different regions. For example, the time of peak water vapor content in southeastern China (D) is one month earlier than other regions, probably due to the well-known rainy season (mei-yu) in southeastern China. In addition, water vapor in spring is a little lower than in autumn over China, which possibly causes a weak semiannual (i.e., 6-month period) variation of PWV. This difference is also consistent with Peixoto et al. (1981), who found during the study of large-scale atmospheric moisture processes that water vapor levels in spring and autumn are not identical but tend to resemble those in their antecedent seasons.

c. Diurnal variation of water vapor

Significant diurnal (24 h) variations of water vapor are found over all GPS stations, and the mean amplitude is about 0.7 mm with an uncertainty of about 0.1 mm (Fig. 10). The higher elevation has a larger amplitude of diurnal variation, with up to about 1.2 mm. The

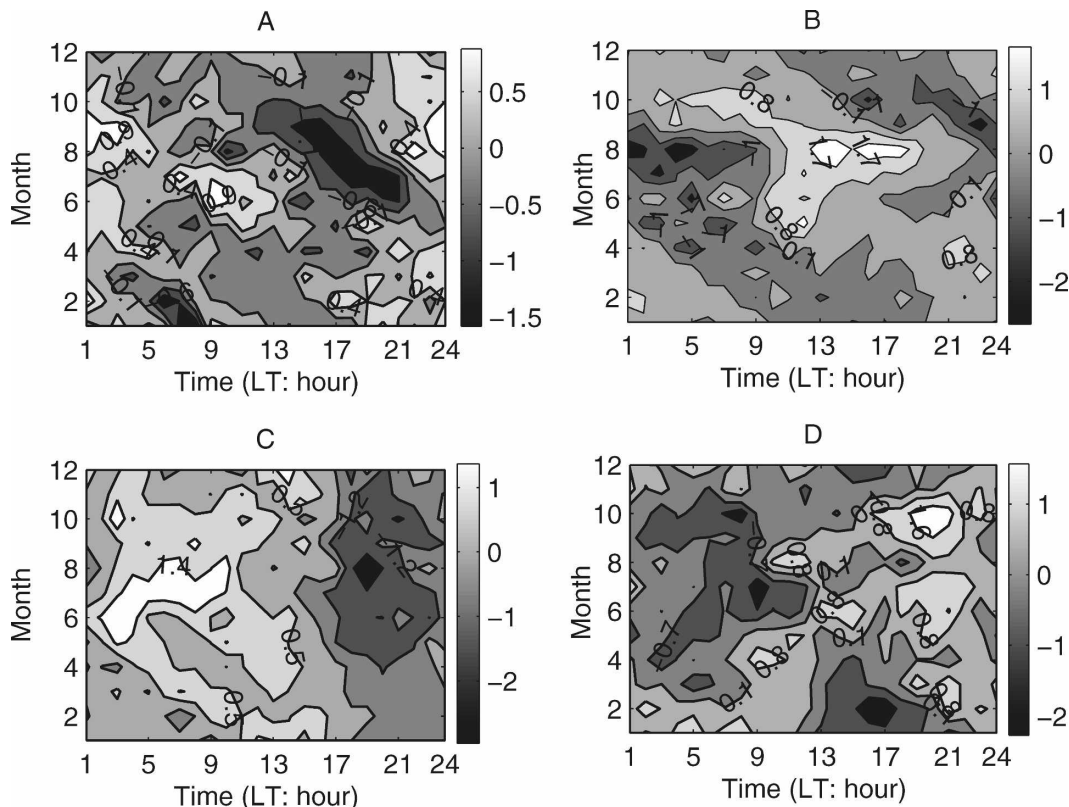


FIG. 12. Mean monthly variations of PWV diurnal cycle anomalies (mm) at the four subregions (A–D) marked in Fig. 1.

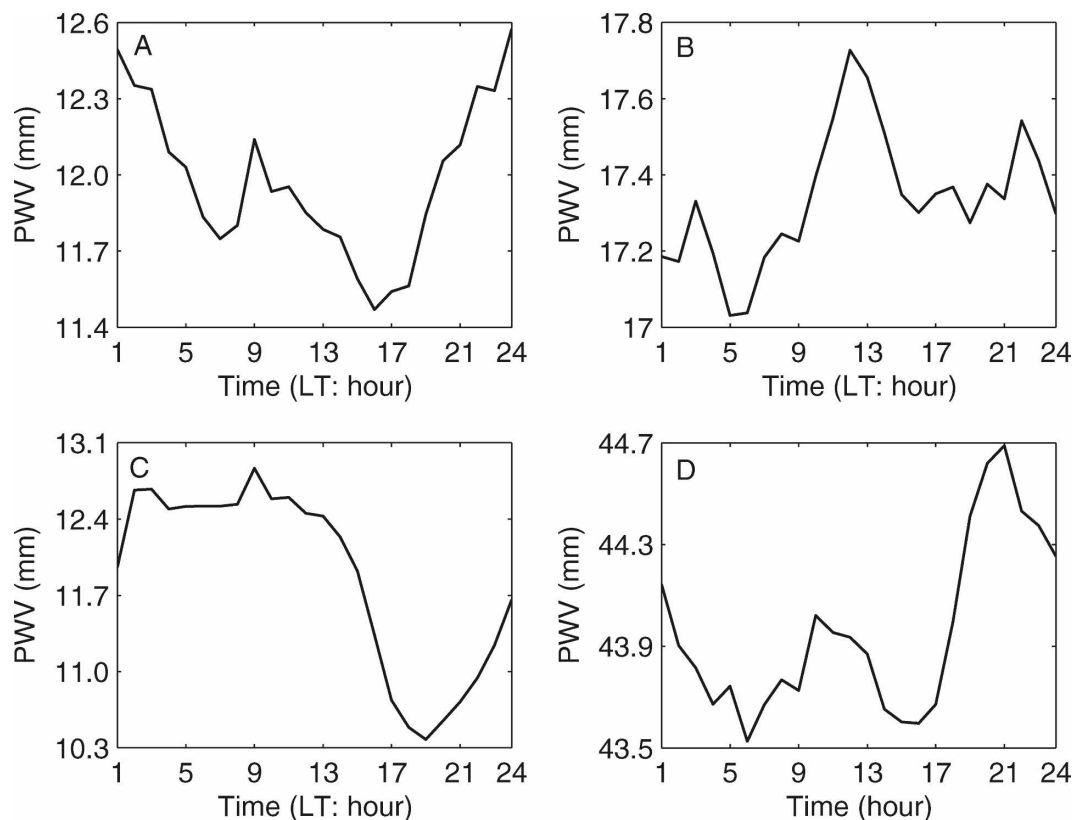


FIG. 13. Mean diurnal variation of PWV at the four subregions (A–D) marked in Fig. 1.

diurnal peak PWV values appear around noon between 0900 and 1300 LT and around midnight between 2000 and 0000 LT, depending on locations, seasons, and so on. The semidiurnal (12 h) cycle is weaker, with an amplitude of normally less than 0.5 mm (excluding one site, Lush, with about 0.7 mm). The larger amplitudes are located at higher elevation areas (Fig. 11). The phase of the semidiurnal cycle is noisier than that of the diurnal variation, and the semidiurnal peak PWV values occur around noon and midnight (for the second cycle). Furthermore, the diurnal and semidiurnal cycles vary with the season. Figure 12 shows the monthly variation of diurnal PWV cycle anomalies at four subregions as marked in Fig. 1. Region C (i.e., Tibet area) has a marked diurnal variation, with the peak time in the morning between 0200 and 1400 LT. The mean diurnal variations of tropospheric water vapor at four subregions are clear to show the diurnal variation features (Fig. 13). The weak semidiurnal oscillations are significant at regions A, B, and D, excluding the winter season (December–February), with a dominant diurnal oscillation. The peak times of semidiurnal oscillations are located in the day between 0900 and 1300 LT and in the night between 2000 and 0000 LT, which may be linked to the diurnal cycle of convection.

4. Conclusions

The 3-yr tropospheric water vapor field is retrieved from continuous GPS observations of the CMONC for 2004–07, which are the first used to investigate the significant variability and distributions over broad and complex geomorphologic China. Distinct seasonal cycles and diurnal variations are found over all GPS stations. The stronger mean water vapors are in southeastern China while northwestern China has lower mean water vapors. The marked seasonal cycles are in summer with a maximum water vapor and in winter with a minimum water vapor. PWV in southeastern China has an annual amplitude of about 15 mm, much larger than in northwestern China with about 4 mm. Meanwhile, the time of peak water vapor content is one month earlier than other regions, probably because of the well-known rainy season (mei-yu). The mean amplitude of significant diurnal variations of water vapor is about 0.7 mm, and the peak diurnal PWV values occur around noon and midnight and are affected by the diurnal cycle of convective activities. The semidiurnal cycle is weaker, with an amplitude of normally less than 0.5 mm and the peak semidiurnal PWV values appear around noon and midnight (for the second

cycle). The spatial distributions and variabilities of PWV are mainly dominated by the latitude, topographical features, the season, and the monsoon. In the future, these PWV datasets will be further applied in numerical prediction models, climatology, and so on.

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